15. Synthesis: The Big Picture, Revisited

Connecting the course, and what comes next

Week 15 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on last week (AI policy and ethics)
2. The map revisited: how every week connects
3. Ten ideas to remember
4. AI literacy in your career
5. What's next: trends, skepticism, and how to keep learning
6. Final presentations: logistics and rubric
7. Course evaluations · Lab day: project dry runs (5-minute practice talks)

Last week in 60 seconds

Week 14: rules, codes, and your responsibility

1 Layers of rules

Laws, state board rules, standards, society codes, organization policies — and your judgment.

2 EU AI Act

Risk tiers; AI used as a safety component of critical infrastructure is high-risk.

3 U.S. patchwork

As of 2026: no single federal AI law — sector regulators, state laws, NIST AI RMF.

4 The seal stays human

The licensed engineer is responsible for sealed work, whatever tools were used.

Learning objectives

By the end of today, you can...

- ✓ Trace how the course's ideas connect, from data to responsibility
- ✓ State ten core ideas about AI, each with an engineering example
- ✓ Describe how to use AI with judgment in internships and early career
- ✓ Evaluate claims about emerging AI trends with healthy skepticism
- ✓ Plan a final presentation that covers every rubric element

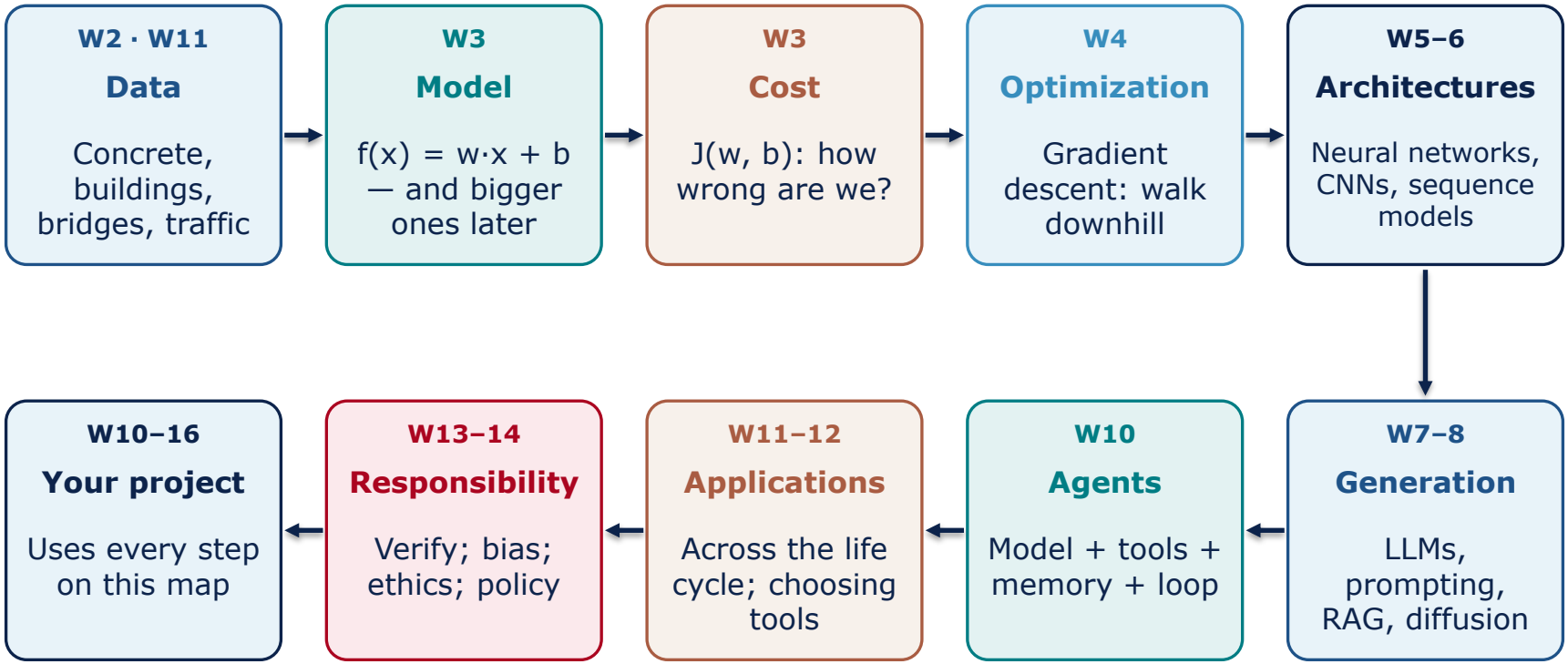
01

The map revisited

How every week connects

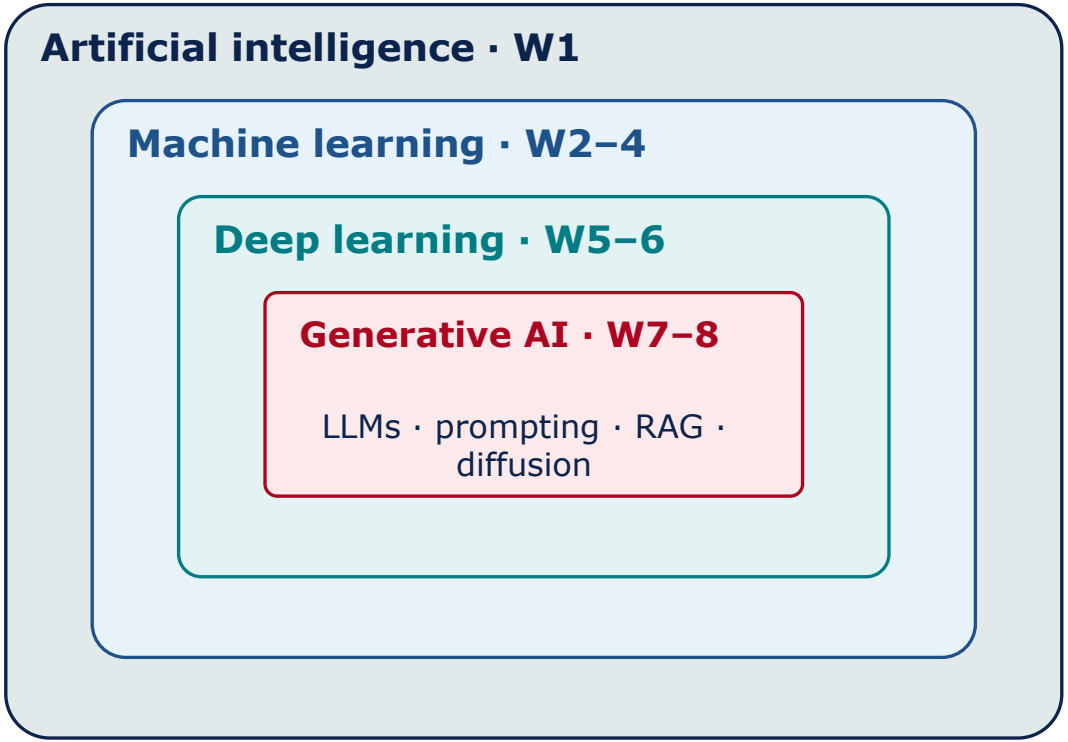
The map revisited

One chain links the whole course



The map revisited

Week 1's map of AI, now filled in



Around and across the map:

Agents · W10
an LLM + tools + memory, in a loop

Beyond ML
optimization, search & planning, rules

Applications · W11–12
every layer, across the life cycle

Responsibility · W13–14
applies to every layer

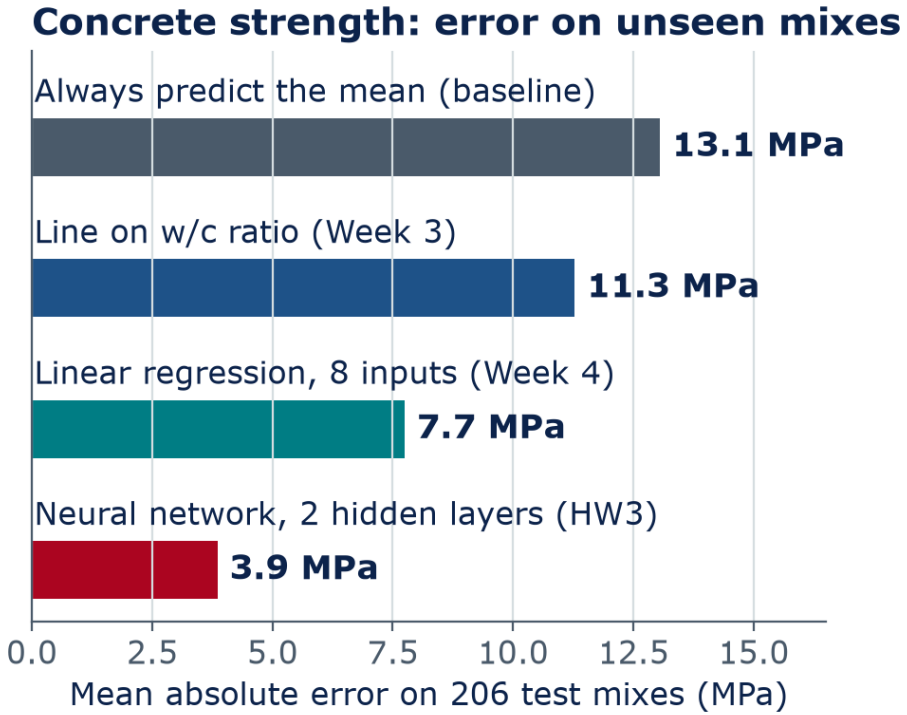
The map revisited

One recipe trained almost everything: model + cost + gradient descent

Model	Weeks	CE/ARCE example	Cost it minimizes	How it learns
Linear regression	3–4	Concrete strength from mix design	Mean squared error	Gradient descent
Logistic regression	4, 11	Is this bridge deck in poor condition?	Cross-entropy (log loss)	Gradient-based optimization
Neural network (MLP)	5	Heating load from building geometry	Mean squared error	Backpropagation + gradient descent
CNN	6	Cracked vs. intact concrete photos	Cross-entropy	Backpropagation + gradient descent
LLM (transformer)	7–8	Next token in an engineering text	Cross-entropy on the next token	Same — then instruction and preference tuning
Agent	10	Look up OSHA rules, run a beam calc	— (no new training)	An LLM in a loop, calling tools

Ideas in action: compare to a baseline, on unseen data

- Same 1,030 concrete mixes from Weeks 2–5
- Every model is scored on **the 206 mixes it never saw**
- No learning (always guess the mean): off by 13 MPa on average
- A small neural network cuts that to about 4 MPa
- The baseline and the test set — not the training error — tell you whether learning worked



The map revisited

Trace your project through the map

5 min

Team huddle

- With your project team, place your project on the map: **data → model → cost/metric → architecture**
- Mark where a **person checks** the result — and where AI tools helped you build it
- Name one risk from Weeks 13–14 that applies to your project
- Two teams share in 30 seconds each

Example (Predict track): Arizona bridge data → logistic regression and random forest → recall on "poor" decks vs. a baseline → an engineer reviews every flagged bridge → risk: some bridge types are rare in the data

02

Ten ideas to remember

If you forget everything else…

Ten ideas to remember

Ideas 1–5: how machines learn

1	**Name the method**	AI is a toolbox. Say which method, trained on what data, for which task.	**W1–2**
2	**Data is a design material**	Quality, quantity, and representativeness limit every model.	**W2, W11**
3	**Model + cost + optimizer**	Guess, measure how wrong, walk downhill. Repeat.	**W3–5**
4	**Test on unseen data**	Beat a baseline. Pick metrics that match the risk.	**W4, W11**
5	**Architecture = assumptions**	CNNs for pixel grids; sequence models for time series.	**W5–6**

Ten ideas to remember

Ideas 6–10: using AI well

6	**LLMs predict plausible text**	Fluent is not the same as correct. Expect hallucinations.	W7
7	**Ground and specify**	Good prompts and retrieval reduce errors — they don't remove them.	W8
8	**Agents act, so limit them**	Few tools, small permissions, step limits, human approval.	W10
9	**Trust, but verify**	Units, magnitudes, independent checks, real citations, tests.	W12
10	**People stay responsible**	Know the failure modes. The engineer — not the tool — signs.	W13–14

Concept check

Which idea applies?

Your team's crack classifier scores 97% accuracy on the test set, but only 3% of the test images show cracks. What should you do first?

A Report it — 97% is excellent

B Compare it to a baseline that always says “no crack,” and check recall on cracked images

C Train it for more epochs to reach 99%

D Add more layers so it can learn more features

Concept check

Which idea applies? — answer

Your team's crack classifier scores 97% accuracy on the test set, but only 3% of the test images show cracks. What should you do first?

A Report it — 97% is excellent

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Always saying "no crack" already scores 97%. Recall on the cracked class shows whether the model finds the cracks that matter (Ideas 2 and 4).

03

AI literacy in your career

Internships, licensure, and judgment

AI literacy in your career

Questions you should be ready to answer

1 "How do you use AI tools?"

Name the tool, the task, and how you checked the output. Use your final project as the example.

2 "When was AI wrong?"

A story from Lab 12 or HW4: what was wrong, how you caught it, what you changed.

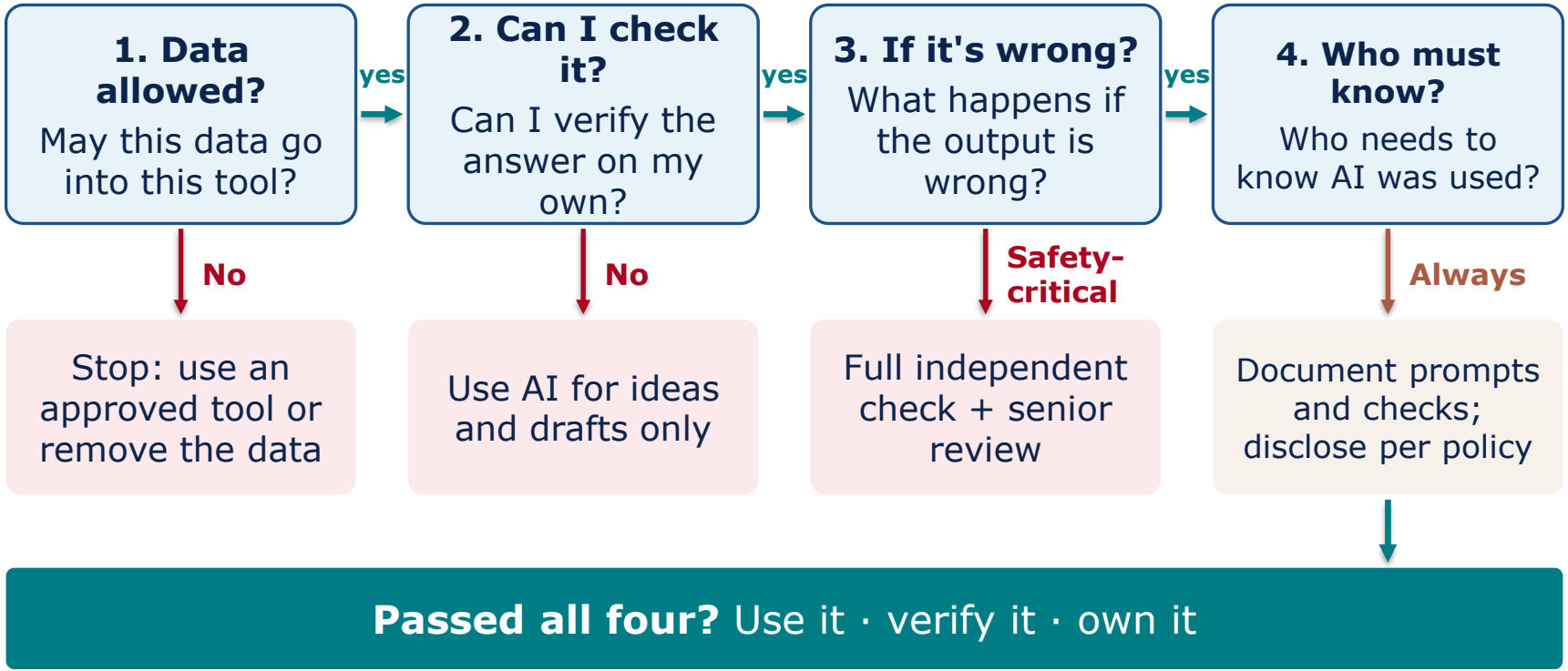
3 "What wouldn't you use it for?"

Unverified safety-critical calcs; client data in public tools; sealing work you didn't check.

4 "Can you work with data?"

Python, pandas, plots, train/test evaluation — on real CE/ARCE data.

Using AI with judgment: four questions first



AI literacy in your career

Your path to licensure — and where AI fits



- Finish the course and project
- Put your project on GitHub

- Ask about the AI policy on day one
- Use approved tools; verify and disclose

- Your own knowledge and an approved calculator
- Then Engineer in Training (EIT)

- Work under a PE's responsible charge
- Check AI output against hand calcs

- You seal the work
- You answer for every tool you used

Draft your 30-second AI-literacy answer

4 min

Think · Pair · Share

- Pick one interview question from two slides ago
- Write a 2–3 sentence answer using a **real example** from this course
- Read it to a partner. Partner asks one follow-up: "How did you check it?"

Template: "In [course/project], I used [method or tool] to [task]. I checked it by [verification]. I learned that [limit or insight]."

04

What's next

Emerging trends — with healthy skepticism

What's next

Four trends to watch — and one question for each

Multimodal agents

- **Promise:** read drawings, photos, and specs, then act in software — draft RFIs (requests for information), update schedules
- **Ask:** what can it change without approval?

AI in BIM

- **Promise:** query building models in plain language; check clashes and code items
- **Ask:** does it work on our models and standards?

Foundation models for science

- **Promise:** e.g., ECMWF has run an ML weather model operationally since 2025
- **Ask:** is it tested on extremes and physically consistent?

Robots on construction sites

- **Promise:** layout printing, rebar tying, drone and legged-robot site scans
- **Ask:** does it work in dust, heat, and clutter, safely near people?

What's next

Signs of hype vs. signs of substance

Signs of hype

- "Fully autonomous," "replaces engineers," "100% accurate"
- A polished demo, but no test data and no baseline
- Silent about failures, data sources, and cost
- No one is accountable when it is wrong

Signs of substance

- A clear task and a clear user
- Results on held-out or real project data, compared to a baseline
- Known failure modes and limits, stated up front
- Human review built in; clear responsibility

What's next

How to keep learning after this course

1 Free courses

- Elements of AI
- Google ML Crash Course
- Hugging Face Learn
- Kaggle Learn

2 Practice

- Redo a lab with new public data: bridges, weather, city open data
- Share it on GitHub

3 Community

- ASCE, AEI, and ASHRAE student chapters
- Hackathons and Kaggle competitions

4 Stay current

- Read primary sources: NIST, standards, professional societies
- Check dates — AI and policy change fast

05

Final presentations

Logistics and the rubric

Final presentations

Finals week: what, when, and how long

10 min

Team presentation during the final-exam period

3–5 min

Questions from the class and the instructor

≤ 6 pages

Final report, submitted with the final notebook

Also due

AI-use statement, reflection, and peer evaluation

≤ 8 teams per 2-h slot; more teams → two rooms or 8-min talks. Schedule on Brightspace (by Week 13).

Final presentations

Where the project grade comes from

The project is 25% of your course grade

Milestone	Weight	Status
Proposal (project canvas)	10%	Done — Week 11
Progress check-in (data + baseline)	10%	Done — Weeks 12–13 labs
Dry run (5-minute practice talk) + peer feedback	5%	This week's lab day (Lab 15)
Final presentation	30%	Finals week: 10 min + 3–5 min Q&A
Final notebook + report (≤ 6 pages)	35%	Finals week; include the AI-use statement
Individual reflection + peer evaluation	10%	Finals week; each person submits

Final presentations

What the rubric looks for: the required elements

Element	What a strong project shows
Problem and users	A specific CE/ARCE question; who would use the answer, and why
Data and its limits	Source, size, cleaning — and what the data cannot represent
Method and baseline	Why this method; a simple baseline to beat
Evaluation	Metrics that match the risk (e.g., recall for rare failures), on test data
Verification	Independent checks: hand calculations, physical sense, known cases
Responsible AI	Risks, bias, failure modes, and who is accountable
AI-use statement	Which AI tools you used, for what, and how you checked their output
Communication	A clear story; readable charts with units; honest limits

Final presentations

A 10-minute talk: the presentation guide's outline

Total 10:00 · about one slide per minute · code and extra charts go in the report

Part	Guide slides	Time	What to show
Problem and question	1–3	1:45	Title; who needs the answer; one measurable question
Data and its limits	4–5	2:00	Source, size, one plot; what the data cannot represent
Method and baseline	6	1:15	Pipeline diagram; the baseline; how the test stayed unseen
Results and failures	7–8	2:30	Baseline vs. your method on test data; where it fails
Verification	9	0:45	The five checks: what you did and what you found
Responsible AI	10	1:00	Biggest risk; who is accountable; needed human review
Takeaway	11	0:45	Use / use with conditions / don't use; what's next

Final presentations

Small changes that make a talk much stronger

Weak	Strong
• "Our model is 95% accurate" — no baseline, no context	• "Recall on poor decks: 0.80, vs. 0.00 for the baseline"
• Slides full of code screenshots	• One clear chart per point, with units
• "We used AI" — with no details	• "We used [tool] to draft plotting code and checked it by …"
• No limits, no failure cases	• Shows where it fails, and who is accountable

vs

Rehearse with a timer — the lab-day dry run is your practice run

Lab 15

Lab day: project dry runs

LAB

15

- Each team gives a **5-minute practice talk** (a "dry run") of the core story of its final talk
- Classmates fill in a structured **peer-feedback form**
- Instructor feedback on content, timing, and slides
- Leave with a list of fixes for finals week
- Bring your slides and notebook — the dry run is 5% of the project grade, not a lab grade

Before you go

Course evaluations: your feedback shapes the next version

1 When

Watch for the university email or Brightspace link, and finish before the deadline.

2 What helps most

Specific comments: which labs, examples, and pacing worked — and what to change.

3 Why it matters

This is a new course. Your comments decide what changes for the next class.

Wrap-up

Key takeaways

1. One chain links the course: data → model → cost → optimization → architectures → generation → agents → applications → responsibility
2. One recipe — model + cost + gradient-based optimization — trains almost everything, from lines to LLMs
3. Judge AI on unseen data, against a baseline, with metrics that fit the risk
4. Use AI with judgment: check the data rules, verify independently, disclose, and **own the result**
5. Meet new trends with curiosity and evidence — and keep learning after this course

Learn more

Readings and resources

- **Elements of AI** — free; revisit the chapters on neural networks and implications
- **Google Machine Learning Crash Course** — free; regression, classification, and fairness modules
- **Hugging Face Learn** — free courses on LLMs and agents
- **Kaggle Learn** — short hands-on Python, pandas, and ML courses
- **fast.ai — Practical Deep Learning** — for those who want to go deeper
- **3Blue1Brown — Neural networks** — visual explanations, including transformers